

3-Axis Magnetic Sensor

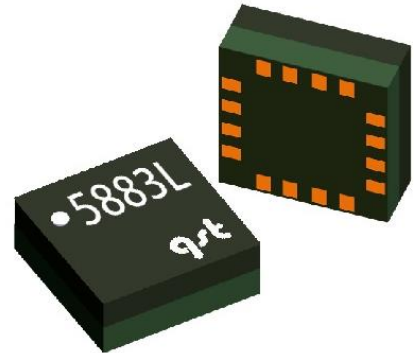
QMC5883L



The QMC5883L is a multi-chip three-axis magnetic sensor. This surface -mount, small sized chip has integrated magnetic sensors with signal condition ASIC, targeted for high precision applications such as compassing, navigation and gaming in drone, robot, mobile and personal hand-held devices.

The QMC5883L is based on our state-of-the-art, high resolution, magneto-resistive technology licensed from Honeywell AMR technology. Along with custom-designed 16-bit ADC ASIC, it offers the advantages of low noise, high accuracy, low power consumption, offset cancellation and temperature compensation. QMC5883L enables 1° to 2° compass heading accuracy. The I²C serial bus allows for easy interface.

The QMC5883L is in a 3x3x0.9mm³ surface mount 16-pin land grid array (LGA) package.



FEATURES

- ▶ 3-Axis Magneto-Resistive Sensors in a 3x3x0.9 mm³ Land Grid Array Package (LGA), guaranteed to operate over an extended temperature range of -40 °C to +85 °C.
- ▶ 16 Bit ADC With Low Noise AMR Sensors Achieves 2 Milli-Gauss Field Resolution
- ▶ Wide Magnetic Field Range (±8 Gauss)
- ▶ Temperature Compensated Data Output and Temperature Output
- ▶ I²C Interface with Standard and Fast Modes.
- ▶ Wide Range Operation Voltage (2.16V To 3.6V) and Low Power Consumption (75µA)
- ▶ Lead Free Package Construction
- ▶ Software And Algorithm Support Available

BENEFIT

- ▶ Small Size for Highly Integrated Products. Signals Have Been Digitized And Calibrated.
- ▶ Enables 1° To 2° Degree Compass Heading Accuracy , Allows for Navigation and LBS Applications
- ▶ Maximizes Sensor's Full Dynamic Range and Resolution
- ▶ Automatically Maintains Sensor's Sensitivity Under Wide Operating Temperature Range
- ▶ High-Speed Interfaces for Fast Data Communications. Maximum 200Hz Data Output Rate
- ▶ Compatible with Battery Powered Applications
- ▶ RoHS Compliance
- ▶ Compassing Heading, Hard Iron, Soft Iron, and Auto Calibration Libraries Available